

Key

Physics 214/224, Quiz 3

Primary instructor: Yurii Maravin

03/25/2022

For grading purposes

Problem 1	
Problem 2	
Problem 3	
Problem 4	
Studio	
Total:	

WID:_____

NAME:_____

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: “On my honor as a student I have neither given nor received unauthorized aid on this academic work.”

For two quick points, circle your studio instructor:

Lohman (TU 7:30) Lohman (TU 9:30) Maravin (TU 11:30)

Kumarappan (TU 1:30) Blaga (FW 9:30) Maravin (FW 11:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.



Part I: All questions to be answered on your Scantron card (34 points total)

1. (2 points) Here is supposed to be a really funny question, but given the fact that it has not passed the EP2 purity law of 1516, it is substituted with the following, a really lame one: who is your EP2 professor?

- ☒ (a) Yurii Maravin
- (b) Vladimir Putin
- (c) Ivan Grozny
- (d) Joseph Stalin (aka Joseph Jugashvili)
- (e) Chinggis Khan

2. (2 points) What appendage do you use in the right hand rule?

- (a) left hand
- ☒ (b) right hand
- (c) left foot
- (d) right eye
- (e) right foot

3. (2 points) What course are we currently enrolled in?

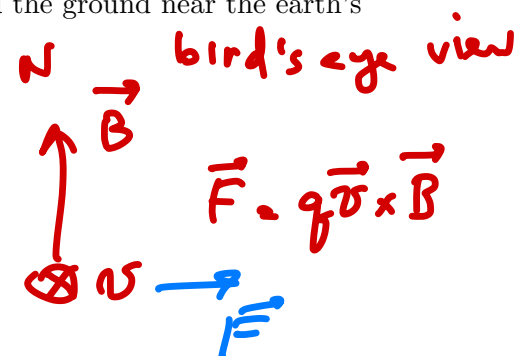
- (a) "South Park" and Contemporary Social Issues
- ☒ (b) Engineering physics II
- (c) Surviving the Coming Zombie Apocalypse – Disasters, Catastrophes, and Human Behavior
- (d) General physics II
- (e) Cryptozoology

4. (2 points) Which alteration below would result in a DC electric motor developing less torque during its operation

- (a) Replacing the motor's aluminum frame with an iron frame
- (b) Reducing the distance between the motor's coil and its magnet's pole faces
- ☒ (c) Halving the area of the motor's coil
- (d) Boosting the current flowing through the motor's coil
- (e) Increasing the number of turns in the motor's coil

5. (2 points) Which way would a proton moving straight down toward the ground near the earth's equator be deflected by earth's magnetic field?

- (a) North
- (b) South
- ☒ (c) East
- (d) West
- (e) Very little deflection will occur



6. (2 points) Which way would an electron moving straight down toward the ground near the center of Antarctica be deflected by earth's magnetic field?

- (a) North
- (b) South
- (c) East
- (d) West
- ☒ (e) Very little deflection will occur

$\vec{B}$ is $\parallel$ to $\vec{v}$

7. (2 points) Magnetic field from the long straight wire

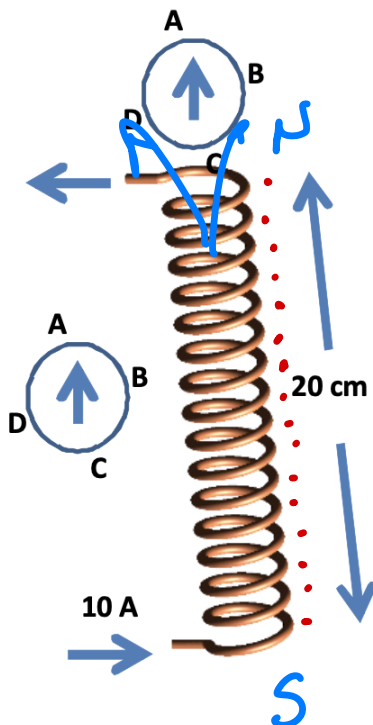
- (a) falls in magnitude with distance as $1/r^2$
- ☒ (b) falls in magnitude with distance as $1/r$
- (c) is constant, just like the electric field from an infinite charged sheet
- (d) increases with distance as r
- (e) increases with distance as r^2

8. (2 points) What is $(1/\mu_0) \times$ the line integral of B around the loop shown below?



- (a) -6 A
- (b) 0 A
- (c) 6 A
- (d) 12 A
- ☒ (e) can be -6 A or 6 A, depending on the choice of the path in the integral.

The next three problems refer to the solenoid depicted below with current flowing in the direction shown.



9. (2 points) The magnetic field at the center of the solenoid is closest to:

- (a) $B = 10^{-7} \text{ T}$
- (b) $B = 10^{-6} \text{ T}$
- (c) $B = 10^{-5} \text{ T}$
- (d) $B = 10^{-4} \text{ T}$
- ☒ (e) $B = 10^{-3} \text{ T}$

$$B = \mu_0 n i = 4\pi \cdot 10^{-7} \frac{\text{Tm}}{\text{A}} \cdot \frac{16}{0.2\text{m}} \cdot 10\text{A}$$

$$= 1.2 \cdot 10^{-6} \cdot 800 \text{ T}$$

$$\approx 1060 \cdot 10^{-6} \text{ T}$$

10. (2 points) The arrow in the center of each circle denotes the north pole of the needle of a compass. Toward which letter will the arrow deflect towards in the top compass when current is flowing in the solenoid?

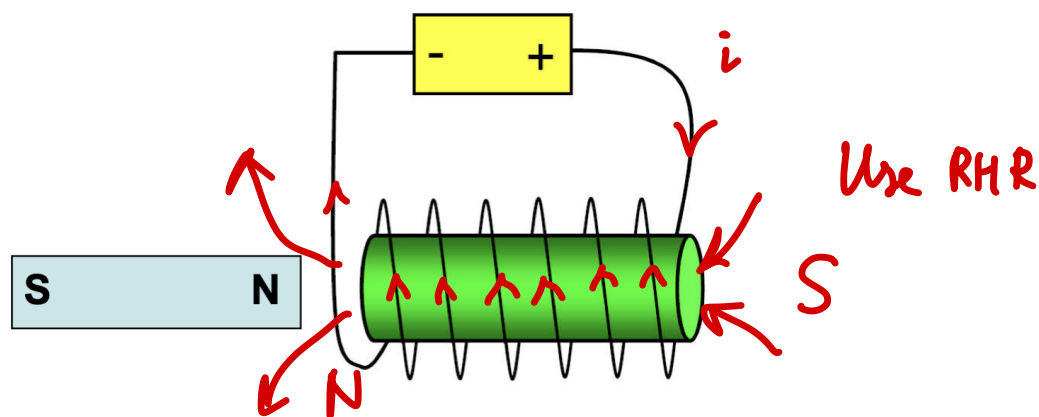
- ☒ (a) A or no deflection
- (b) B
- (c) C
- (d) D

use RHR

11. (2 points) The arrow in the center of each circle denotes the north pole of the needle of a compass. Toward which letter will the arrow deflect towards in the compass on the left side of the solenoid, when the current is flowing?

- ☒ (a) A or no deflection
- (b) B
- (c) C
- (d) D

12. (2 points) What will happen to the magnet in the figure below if the current starts to flow through the solenoid?



- (a) nothing
 (b) the magnet will be pushed to the left
 (c) the magnet will be pushed to the right
 (d) the magnet will be pushed up
 (e) the magnet will be pushed down
13. (2 points) The magnetic field produced by a magnetic dipole falls off in strength as $1/\text{distance}^3$ for points far away from the dipole. If a particular small coil produces a 0.1 T field at a distance of 1 m directly above its center, what will the field be 10 m directly above the center?

- (a) 10^{-2} T
 (b) 10^{-3} T
 (c) 10^{-4} T
 (d) 10^{-5} T
 (e) 10^{-6} T

it will be $\frac{1}{10^3}$ smaller $\Rightarrow$
 $0.1 \text{ T} \cdot 10^{-3} = 10^{-4} \text{ T}$

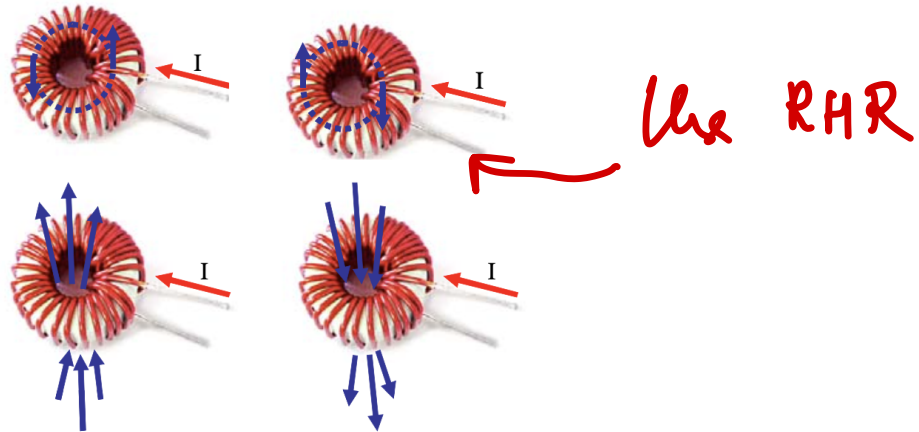
14. (2 points) A proton and an α -particle ($m_\alpha/m_p = 4$, $Q_\alpha/Q_p = 2$) have the same momentum and move in a constant B -field at right angles to $\vec{B}$. How will the radius of the α -particle circle compare to that of the proton?

- (a) $R_\alpha/R_p = 4$
 (b) $R_\alpha/R_p = 2$
 (c) $R_\alpha/R_p = 1$
 (d) $R_\alpha/R_p = 0.5$
 (e) $R_\alpha/R_p = 0.25$

$$\frac{mv^2}{R} = qvB \quad R = \frac{mv}{qB} = \frac{p}{qB}$$

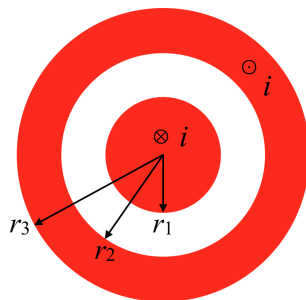
$$\frac{R_\alpha}{R_p} = \frac{q_p p}{q_\alpha p} = \frac{1}{2}$$

15. (2 points) The toroid above receives its current in the direction shown. Which field line map (arrows) best describes the shape of the magnetic field in the toroid?



- (a) Upper left
 - ☒ (b) Upper right
 - (c) Lower left
 - (d) Lower right
16. (2 points) A piece of the element gadolinium is placed inside a solenoid and subject to a 0.2 T magnetic field. After the gadolinium is removed from the solenoid, it can still deflect the needle of a compass. From this, we conclude that gadolinium is
- (a) Non-magnetic
 - (b) Diamagnetic
 - (c) Paramagnetic
 - ☒ (d) Ferromagnetic
 - (e) No conclusions are possible
17. (2 points) A piece of platinum is weakly attracted by either pole of a strong bar magnet. From this we conclude that platinum is
- (a) Non-magnetic
 - (b) Diamagnetic
 - ☒ (c) Paramagnetic
 - (d) Ferromagnetic
 - (e) No conclusions are possible

Part II: Work out this part in the space provided. Show your work!
(39 points total)



18. (9 points) In a very long coaxial cable, an inner conducting wire of radius $r_1 = 1.00$ mm carries a current $i = 10$ mA into the page as shown. An outer conducting shield of inner radius $r_2 = 1.5$ mm and outer radius $r_3 = 2$ mm carries an equal current of 10 mA in the opposite direction. Assume the current density is uniform. Calculate the magnitude of the magnetic field in units of Tesla, at the following radii:

- (a) (2 points) $r = 0$ mm, i.e., at the center of the inner wire.

Due to radial symmetry $\vec{B}$ must be zero
Explanation (+1)

Do not award point for Ampere's law as $r=0$ and it's not valid, unless they take a limit.

- (b) (3 points) $r = 0.5$ mm.

This is within a central wire, so use Ampere's Law:

$$\oint \vec{B} d\vec{\ell} = \mu_0 i_{\text{enc}} \Rightarrow 2\pi r B = \mu_0 i_{\text{enc}} \Rightarrow B = \frac{\mu_0 i_{\text{enc}}}{2\pi r} \quad (+1)$$

To find i_{enc} , use current density in the central wire

$$j = \frac{i}{\pi r_1^2} \Rightarrow i_{\text{enc}} = j \cdot \pi r^2 = \frac{i}{\pi r_1^2} \pi r^2 = \frac{i r^2}{r_1^2} \quad (+1) \Rightarrow B = \frac{\mu_0 i r^2}{2\pi r_1^2 r} \Rightarrow$$

$$B = \frac{\mu_0 i r}{2\pi r_1^2} = \frac{4\pi \cdot 10^{-7} \frac{\text{Tm}}{\text{A}} \cdot 10 \cdot 10^{-3} \text{ A} \cdot 5 \cdot 10^{-4} \text{ m}}{2\pi \cdot (10^{-3} \text{ m})^2} = 10^{-6} \text{ T} = 1 \mu\text{T} \quad (+1)$$

- (c) (2 points) $r = 1.25$ mm.

This is in the cavity, between the shell and inner wire, so

$i_{\text{enc}} = 10 \text{ mA}$ (+1) and from Ampere's law:

$$2\pi r B = \mu_0 i_{\text{enc}} \Rightarrow B = \frac{\mu_0 i_{\text{enc}}}{2\pi r} = \frac{4\pi \cdot 10^{-7} \frac{\text{Tm}}{\text{A}} \cdot 10 \cdot 10^{-3} \text{ A}}{2\pi \cdot 1.25 \cdot 10^{-3} \text{ m}} =$$

$$= 1.6 \cdot 10^{-6} \text{ T} = 1.6 \mu\text{T} \quad (+1)$$

- (d) (2 points) $r = 2.5$ mm.

This is outside the coaxial cable, so the enclosed current is the sum (with proper signs) of all the currents:

$$(+1) i_{\text{enc}} = 10 \text{ mA} - 10 \text{ mA} = 0 \Rightarrow B = 0 \text{ T} \quad (+1) \text{ from the Ampere's law.}$$



19. (7 points) Arnold Schwarzenegger urgently needs a railgun to save the world. Your job is to help him with a design using your extensive knowledge of EP2 material. Assume that the railgun should shoot 10 gram tungsten slugs 5 cm long and 1 cm wide, each having a resistance of about $1 \mu\Omega$ across the diameter, where the current is flowing when the railgun is fired.

- (a) (2 points) Assuming that Arnold is strong enough to carry a power supply that provides a constant 400 V power source, estimate the current flowing through the slug when the power supply is shorted on a slug upon firing.

From Ohm's law:
$$i = \frac{V}{R} = \frac{400V}{1\mu\Omega} = 400 \cdot 10^6 A = 4 \cdot 10^8 A$$

We ignore other resistance in the circuit

- (b) (2 point) It is very important for Arnold that the railgun round goes toward the enemy when he fires the rifle. If the current flows through the slug from Arnold's left to right, which way does the magnetic field have to point to ensure that the gauss round would fire from the muzzle?

view from top

Left right

$$\vec{F} = i \vec{L} \times \vec{B}$$
, we RHR (+1)
 $\vec{B}$ must point down (+1)

- (c) (3 points) Estimate the magnetic field needed to accelerate the slug to about 1000g.

(+1)
$$\vec{F} = iL\vec{B} = ma$$
 (+1)

$$B = \frac{ma}{iL} = \frac{10 \cdot 10^{-3} \text{ kg} \cdot 1000 \cdot 9.8 \text{ m/s}^2}{4 \cdot 10^8 \text{ A} \cdot 10^{-2} \text{ m}} = 2.45 \cdot 10^{-5} \text{ T}$$

$$B = 0.245 \mu\text{T}$$
 (+1)

20. (9 points) An ion with a net charge of $+e$ is tracked through a mass spectrometer operating with a magnetic field $B = 0.5 \text{ T}$ that is oriented at 90 degrees to the ion's momentum vector. The ion is observed to follow a semicircular path of radius $R = 0.2 \text{ m}$ that requires $t = 2.4 \times 10^{-6} \text{ s}$ to complete.

(a) (3 points) Determine the momentum of the ion.

$$\frac{mv^2}{R} = qvB \Rightarrow \frac{mv}{R} = qB \Rightarrow \frac{p}{R} = qB$$

$$p = qBR = 1.6 \cdot 10^{-19} \text{ C} \cdot 0.5 \text{ T} \cdot 0.2 \text{ m} = 1.6 \cdot 10^{-20} \frac{\text{kg m}}{\text{s}}$$

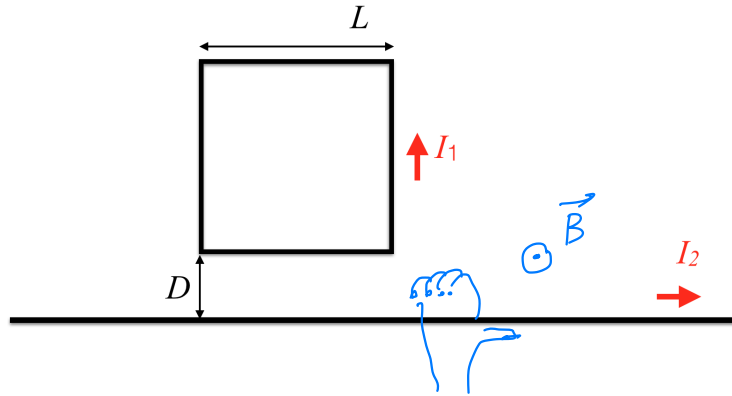
(b) (3 points) Determine the speed of the ion.

The semicircular path has a length πR and it takes t time to cover $\Rightarrow v = \frac{\pi R}{t} =$

$$= \frac{\pi \cdot 0.2 \text{ m}}{2.4 \cdot 10^{-6} \text{ s}} \approx 2.61 \cdot 10^5 \text{ m/s}$$

(c) (3 points) Determine the mass of the ion.

$$p = mv \Rightarrow m = \frac{p}{v} = \frac{1.6 \cdot 10^{-20} \frac{\text{kg m}}{\text{s}}}{2.61 \cdot 10^5 \text{ m/s}} = 6.1 \cdot 10^{-26} \text{ kg}$$



21. (14 points) A square loop with a side size L carries a current I_1 in a counter-clockwise direction, as shown on picture above. The loop is positioned at a distance D from a very long wire carrying a current I_2 from left to right.

- (a) (3 points) Calculate the magnetic field from the long wire at any point that is separated from the wire at a distance r . What is the direction of the magnetic field above the wire?

(+1) Use Ampere's law: $2\pi r B = \mu_0 i_2 \Rightarrow B = \frac{\mu_0 i_2}{2\pi r}$ (+1)

Use RHR: $\vec{B}$ points toward the viewer. (+1)

- (b) (3 points) What is the magnetic dipole moment of the loop of the wire?

$\vec{\mu}_B = i \vec{A} = i_1 L^2$ (+2) (+1)

- (c) (4 points) What is the magnitude and the direction of the net force on the loop exerted by the wire? Assume $D = 100$ m, $L = 2$ m, $I_1 = 1$ A, $I_2 = 20$ A.

(+1) $\vec{F} = i \vec{L} \times \vec{B}$. Sides have forces that cancel due to symmetry, only top/bottom remain $\Rightarrow$ (+1)

(+1) $F_{\text{net}} = i_1 L \left(\frac{\mu_0 i_2}{2\pi D} - \frac{\mu_0 i_2}{2\pi(D+L)} \right) = \frac{\mu_0 i_1 i_2 L}{2\pi D(D+L)} \cdot (D+L - D) = \frac{\mu_0 i_1 i_2 L^2}{2\pi D(D+L)}$ (+1)

- (d) (4 points) Use the formulae that you obtained in (a) and (c) to calculate a force on the loop, using the approximation $F = \left| \frac{dB}{dr} \right| \mu_B$, where μ_B is the magnetic dipole moment of the loop. Take the value of the $\left| \frac{dB}{dr} \right|$ at r corresponding to the center of the loop. How does the answer compares with what you obtained in (c)?

$\frac{dB}{dr} = \frac{d}{dr} \left(\frac{\mu_0 i_2}{2\pi r} \right) = - \frac{\mu_0 i_2}{2\pi r^2} \Rightarrow F = \frac{\mu_0 i_2}{2\pi r^2} \cdot i_1 L^2 =$ (+2) (+1)

$= \frac{\mu_0 i_1 i_2 L^2}{2\pi D^2} = 1.60 \cdot 10^{-9} \text{ N}$ (+1) very close!